ENV S 193DS

Final

Liv Zou

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Setup chunk:

```
library(tidyverse)
library(dplyr)
library(janitor)
library(here)
```

Problem 1

1a.

He most likely used a logistic regression because they were testing whether the distant to water edge predicted for the probability of microhabitat use. In the second part, he most likely perform a chi square test of independence because they were testing whether bird species and habitat (which are categorical variables) were related.

1b.

Part A:

The American Avocets were more or less likely to use a microhabitat depending on their distance from the water. Distance to water's edge is a significant factor that influences microhabitat use.

Part B:

Use of habitat showed to be different across Avocets, Forster's Terns, and Black-necked Stilts. The three species showed unique patterns in the use of managed ponds, marshes, salt ponds, and mudflats.

1c.

An additional variable that could contribute to the probability of Avocet microhabitat use is human activity. Maybe it could be measured as a categorical variable organized in low, medium, and high levels, assessed by how many people are present in the habitat hourly. The birds may avoid those areas with more human activity because it might affect how much the birds go out of their shelters/nests to go eat, or may trigger them to fly out of their habitat when people are really close to them.

Problem 2

2a.

```
nests <- read_csv(here("data", "nest_data_final.csv"))
```

```
# Clean dataset
nests_clean <- nests |>
  # Select relevant columns according to metadata
     select(ant_species, height_cm, bird_species) |>
  # Select the relevant ant species
filter(ant_species %in% c("AB", "RRB", "BBR")) |>
mutate(
  # Reorder ant species
  # Set desired order
    ant_species = factor(ant_species,
                         levels = c("AB", "RRB", "BBR")),
    # Creating new columns displaying ant names
    scientific_name = case_when(
      ant_species == "AB"  ~ "C. sjostedti",
      ant_species == "RRB" ~ "C. mimosae",
      ant_species == "BBR" ~ "C.nigriceps"))
```

```
# Convert data into string and display structure
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ ant_species    : Factor w/ 3 levels "AB","RRB","BBR": 2 2 2 2 1 2 3 2 1 3 ...
 $ height_cm      : num [1:270] 392 310 513 154 324 ...
 $ bird_species   : chr [1:270] "GHS" "GHS" "GHS" "GHS" ...
 $ scientific_name: chr [1:270] "C. mimosae" "C. mimosae" "C. mimosae" "C. mimosae" ...
```

```r
# Take 10 random samples
slice_sample(nests_clean, n = 10)
```

```
# A tibble: 10 x 4
   ant_species height_cm bird_species scientific_name
   <fct>           <dbl> <chr>        <chr>
 1 RRB              180  GCSW         C. mimosae
 2 RRB              398. GHS          C. mimosae
 3 RRB              420. GHS          C. mimosae
 4 RRB              204  STAR         C. mimosae
 5 RRB              170. STAR         C. mimosae
 6 RRB              134. GHS          C. mimosae
 7 AB               114. STAR         C. sjostedti
 8 RRB              110. GCSW         C. mimosae
 9 RRB              204. STAR         C. mimosae
10 RRB              196. GCSW         C. mimosae
```

2b.

In the case column, a value of 1 means that a bird nest was present in the tree habitat while 0 indicates the absence of a bird nest. The values 1 and 0 represent if a tree was used as a nesting site or not.

2c.

The predictor variables are tree height and acacia ant species.

Tree height is a continuous quantitative variable, because it can be measured. As the height increases the probability of more nests appearing is expected to increase because the trees get bigger and taller, providing more spaces for birds to form shelteres that are not in the proximity of any predators that could probably harm them. This information is supported in the paper's results section where the probability of a bird deciding to nest increased when the trees increased by a 1 meter interval.

Ant species is a categorical variable, because it has unique categories of something, in this case, there are distinctive, unique species of ants. Ant species may also influence the probability of nest appearance, and the abstract and introduction of Lujan's paper describes the relationship of ants and birds to be mutualistic.

2d.

model_number	model_description	predictors	interaction_term
1	Nest occurrence predicted by height and species without interactions	height_cm + ant_species	Yes
2	Nest occurrence predicted by height species with interaction btwn variables.	height_cm * ant_species	No

2e.

```
table <- tibble(
  model_number = c(1, 2),
  model_description = c(
    "Nest occurrence predicted by height and species without interactions",
"Nest occurrence predicted by height and species w/ interaction btwn variables."
  ),
  predictors = c(
    "height_cm + ant_species",
    "height_cm * ant_species"
  ),
  interaction_term = c("Yes", "No")
)
```